

Internship Report

Validation of the Porosity-Turbulence Model in TrioCFD

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Chapter 1

Introduction

This report summarizes the work done so far on the validation of the porosity-turbulence model implemented in the CFD code **TrioCFD**. This model, developed as part of Antonin Couvez's PhD, aims to represent turbulent flow behavior inside porous media such as clogging layers on the tube support plates of steam generators.

The goal is to validate the model against reference cases by reproducing the same physical behavior (velocity, pressure profiles) using the porous model with varying porosity. A range of configurations were simulated and compared.

1.1 Overview of the Model

The model modifies the RANS equations to account for porosity and adds source terms depending on two empirical constants c_0 and c_1 .

$$\begin{aligned} C_p &= c_0 \left(0.5(1 - \phi) + \left(1 - \frac{1}{\phi} \right)^{2.2} \right) \\ K &= 0.063 c_1 \frac{\phi^4}{1 - \phi} L^2 \\ \frac{\alpha}{\sqrt{K}} &= \frac{0.774}{\sqrt{c_1} L} \frac{1 - \phi}{\phi^{3.75}} \end{aligned} \tag{1.1}$$

These constants are:

- c_0 , which affects momentum loss (pressure drop) through the C_p term.
- c_1 , that related to the permeability K and the inertial resistance α .

The porous region is defined by a porosity value ϕ . As $\phi \rightarrow 0$, the medium behaves like a fluid, while for $\phi \rightarrow 1$, the medium behaves like a solid.

Simulation Setup

The model equations were implemented using the **Baltik** interface in TrioCFD. Two configurations are used:

- **Reference case:** The solid zone is created by removing parts of the mesh. The parameter L_x represents the width of this solid zone. In other words, if $\phi = 0$, then $L_x = 0$ (i.e., no solid region). As ϕ increases towards 1, L_x increases towards a constant value, typically chosen as one-third of the total channel width. This setup ensures symmetry: equal solid zones on the left and right, and a central opening for the fluid to flow through.

- **Validation case:** The same zone is modeled as a porous region using the implemented porosity model; i.e., no part of the mesh is removed.

Simulations are run using automated scripts in a Jupyter Notebook that handle input generation, launching the runs, and comparing the outputs.

Error Metrics

We evaluate error using relative norms:

- L_1 : Mean relative error
- L_2 : Root mean square relative error
- L_∞ : Maximum relative error

These are computed for both pressure and velocity profiles.

Chapter 2

Case 1

2.1 Finding the Parameters

A wide range of c_0 and c_1 values were tested over $\phi \in [0.1, 0.9]$:

- At first I started with

$$c_0 = [0.003, 0.01, 0.1, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 2.0], \quad c_1 = [1.0, 3.2, 5.0, 10, 20]$$

Why these values? No particular reason, just had to choose somewhere to start (in hindsight it wasn't a great strategy). I looked at the error for V_{haut} (top of the channel at $y = 0.006$ m), and noticed that the error was decreasing while decreasing c_0 and increasing c_1 . So naturally I kept the same c_0 range and increased c_1 : first up to 50, then 100, then 1000. Same observation.

- Then I decided to decrease c_0 as much as possible and increase c_1 as much as possible, and also look at the pressure error. So I tested:

$$c_0 = [10^{-9}, 10^{-8}, 10^{-7}, 10^{-6}, 10^{-5}], \quad c_1 = [10^4, 10^5, 10^6, 10^7, 10^8, 10^9]$$

Naturally, the best error was at the lowest $c_0 = 10^{-9}$ and highest $c_1 = 10^9$.

- However, a problem emerged when looking at the velocity profile in VisIt: there were very large peaks at the interface of the porous region; which is physically very unnatural. After discussing with Antonin, we decided to change the strategy and instead optimize for **pressure loss**.
- Since c_0 is the constant that controls the pressure loss (see the equation for C_p), the idea was to fix $c_0 = 0$ (or something close to 0), and then try to find the best c_1 that minimizes the pressure loss.

2.1.1 Finding c_1

Due to code limitations, the smallest c_0 I could run with was 10^{-6} ; so I used this as my “zero”. I ran a new batch with:

$$c_1 \in [10^3, 10^4, 10^5, 10^6, 10^7, 10^8]$$

To save computation time, I only used 3 values of $\phi = 0.3, 0.5, 0.7$, because after running all the previous simulations I didn't notice any major difference across the ϕ values from 0.1 to 0.9. So I believe I'm justified in using only 3 values.

After some trial and error the optimal value for c_1 was somewhere in the range 10^4 – 10^5 (see Figure 2.1). Having to choose one value meant trying to balance the errors between the different norms, especially L_1 and L_2 ; thus $c_1 = 10^4$ was chosen.

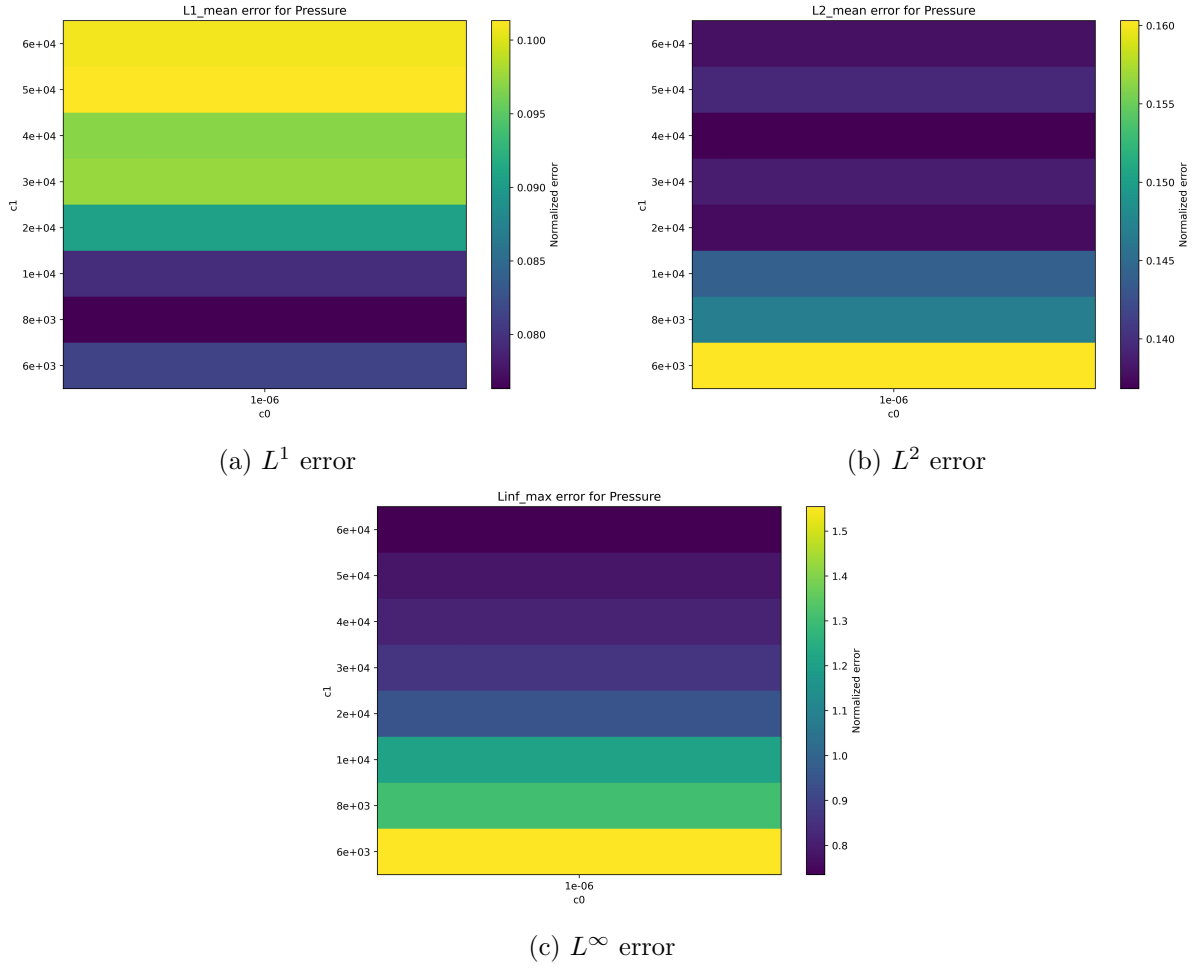


Figure 2.1: Pressure loss error distribution heatmaps for c_1 between 3×10^3 and 6×10^4 : (a) L^1 , (b) L^2 , (c) L^∞ .

2.1.2 Finding c_0

I fixed $c_1 = 10^4$ and searched for a good c_0 value (mostly in the $[0, 1]$ range as discussed with Antonin). After testing the following:

$$c_0 = [10^{-6}, 10^{-3}, 10^{-2}, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0]$$

Figure 2.2 shows the error distribution for pressure loss; for the same reasons as before I found a good compromise at:

$$c_0 = 0.4$$

But what about the velocity?

When we fix c_0 at 10^{-6} , we're only minimizing the pressure loss, which gives us $c_1 = 10^4$ as the definite best value. But if we want to find c_0 while using this c_1 , we need to decide what to focus on; should it be pressure, velocity, or something in between?

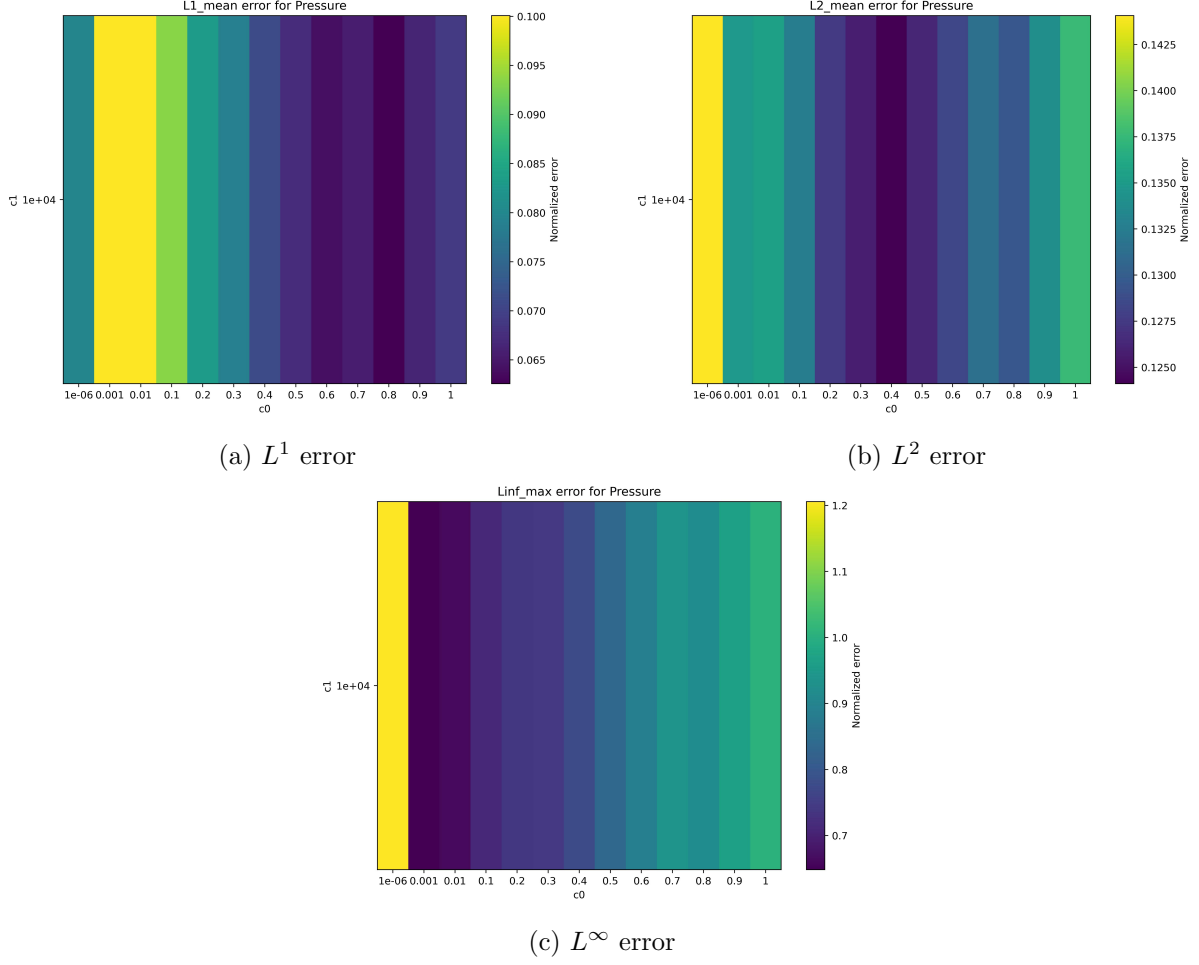


Figure 2.2: Pressure loss error distribution heatmaps for c_0 between 10^{-6} and 1: (a) L^1 , (b) L^2 , (c) L^∞ .

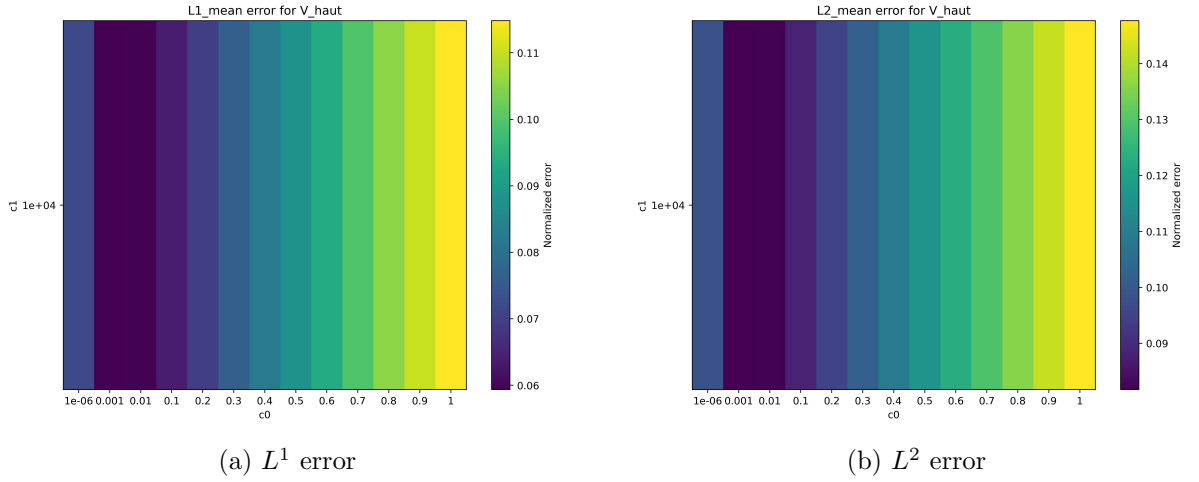


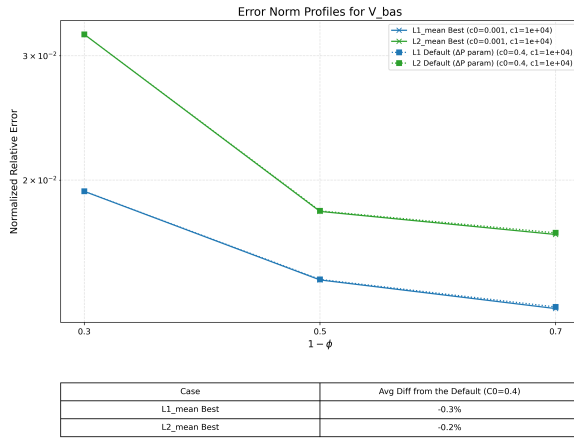
Figure 2.3: Velocity at the top of the channel; error distribution heatmaps for c_0 between 10^{-6} and 1: (a) L^1 , (b) L^2 .

For example, if we look at the heatmap for the velocity at the top of the channel (Figure 2.3), we can see that the best value for c_0 is in the 0.001–0.01 range. For this part, I'm focusing on L_1 and L_2 errors (not L_∞), looking for values that work reasonably well for both rather

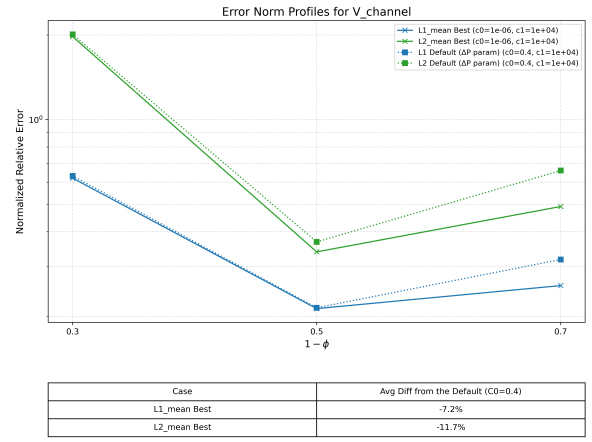
than optimizing for either one individually. When analyzing pressure loss for $c_1 = 10^4$ across $c_0 \in [0, 1]$, $c_0 = 0.4$ comes out as the best choice. So the combination $(c_0 = 0.4, c_1 = 10^4)$ is optimal for pressure loss; but does it hold up when we look at velocity profiles?

To check this, we plot the best parameter combinations for each velocity probe against our default pressure-optimized pair (see graphs below). Since the "best" parameters change depending on which velocity we're looking at, we need to see if sticking with our pressure-derived values is justified. We examine four profiles:

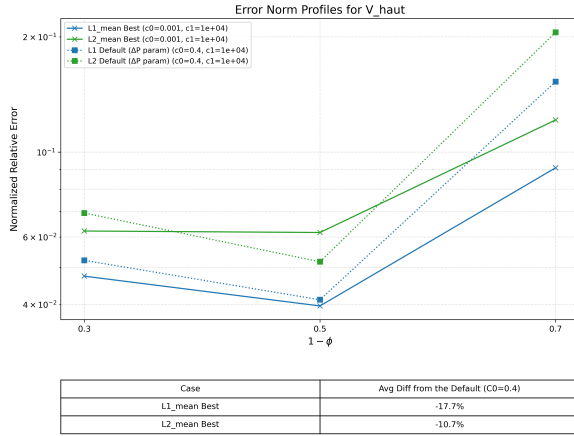
- Bottom horizontal profile
- Channel center profile
- Top horizontal profile
- Vertical profile (full channel height)



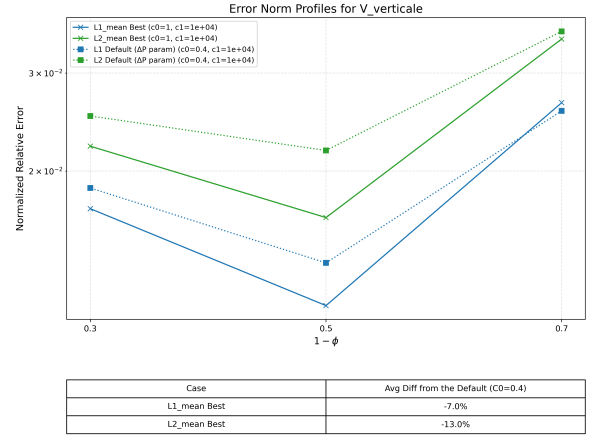
(a) Bottom horizontal profile



(b) Channel center profile



(c) Top horizontal profile



(d) Vertical profile

Figure 2.4: Error profiles comparing default (dashed) and optimized (solid lines) parameters.

From the graphs (Figure 2.4), when we compare the optimal parameters for each probe against our default ($c_0 = 0.4, c_1 = 10^4$); and if we ignore V_{bas} (no significant difference) and average the rest, we're only giving up, on average, about 11% accuracy by using the default values.

Is this trade-off worth it? I'd say yes. We're simplifying our setup significantly by fixing these parameters, and in return we're getting decent performance across all cases. Could we

squeeze out slightly better results by tweaking parameters for each specific case? Probably. But for practical purposes, $c_0 = 0.4$ and $c_1 = 10^4$ give us good enough results.

It's worth noting that generally, for the velocities probes the best errors came with low values of c_0 in the range of 10^{-6} - 10^{-2} .

2.2 Analysis of Case 1 with $c_0 = 0.4$, $c_1 = 10^4$

Now that we have identified the best parameters, I will show a more comprehensive analysis using a refined mesh and a wider range of porosities ($\phi \in [0.2, 0.9]$). The meshes used for the reference and validation cases are shown in Figure 2.5.

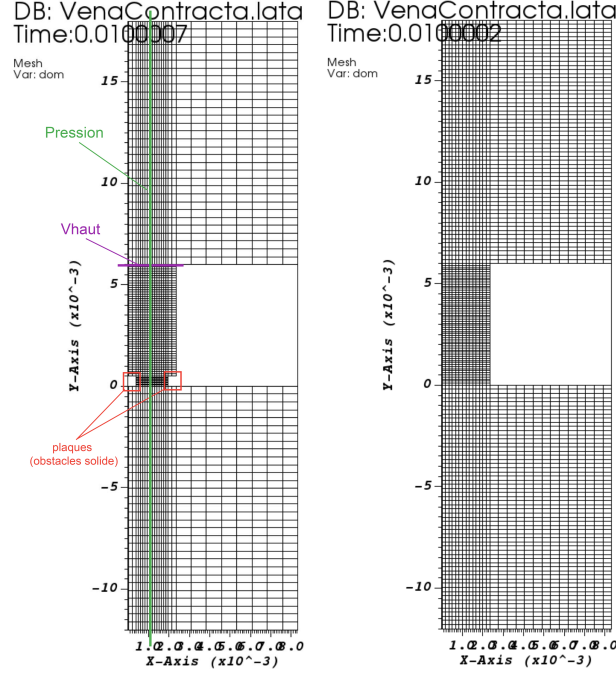
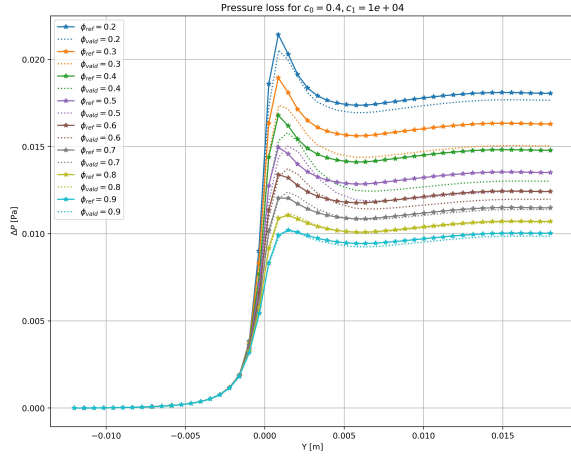


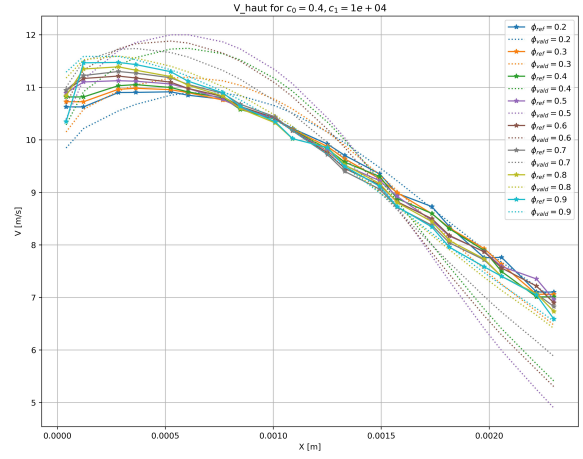
Figure 2.5: Meshes used for the refined simulations: reference case (left) and validation case (right).

We now examine the various probes of interest over the range of porosities $0.1 \rightarrow 0.9$. Figure 2.6 shows the comparison between reference and validation cases for:

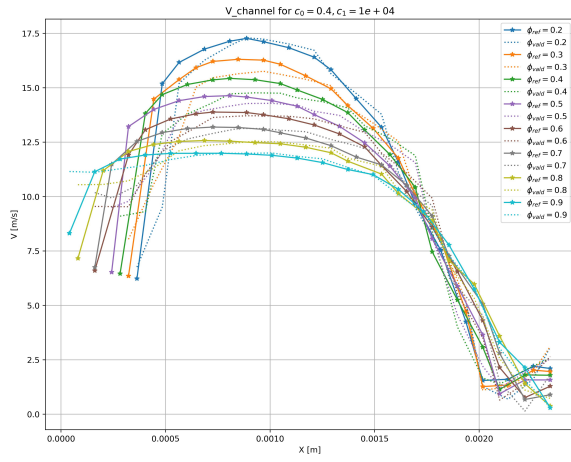
- Pressure loss
- Velocity at the top of the channel (V_{haut})
- Velocity at the channel center (V_{channel})
- Velocity at the bottom of the channel (V_{bas})



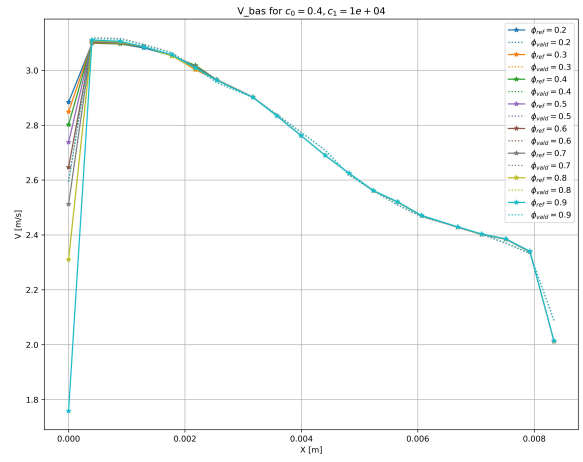
(a) Pressure loss profile



(b) V_{haut}



(c) V_{channel}



(d) V_{bas}

Figure 2.6: Profiles from different probes for the reference and validation cases over $\phi \in [0.2, 0.9]$ using $c_0 = 0.4$, $c_1 = 10^4$.

We also look at the L -norm errors for these simulations (Figure 2.7), using the same probes. This allows us to quantify the differences between the reference and validation cases.

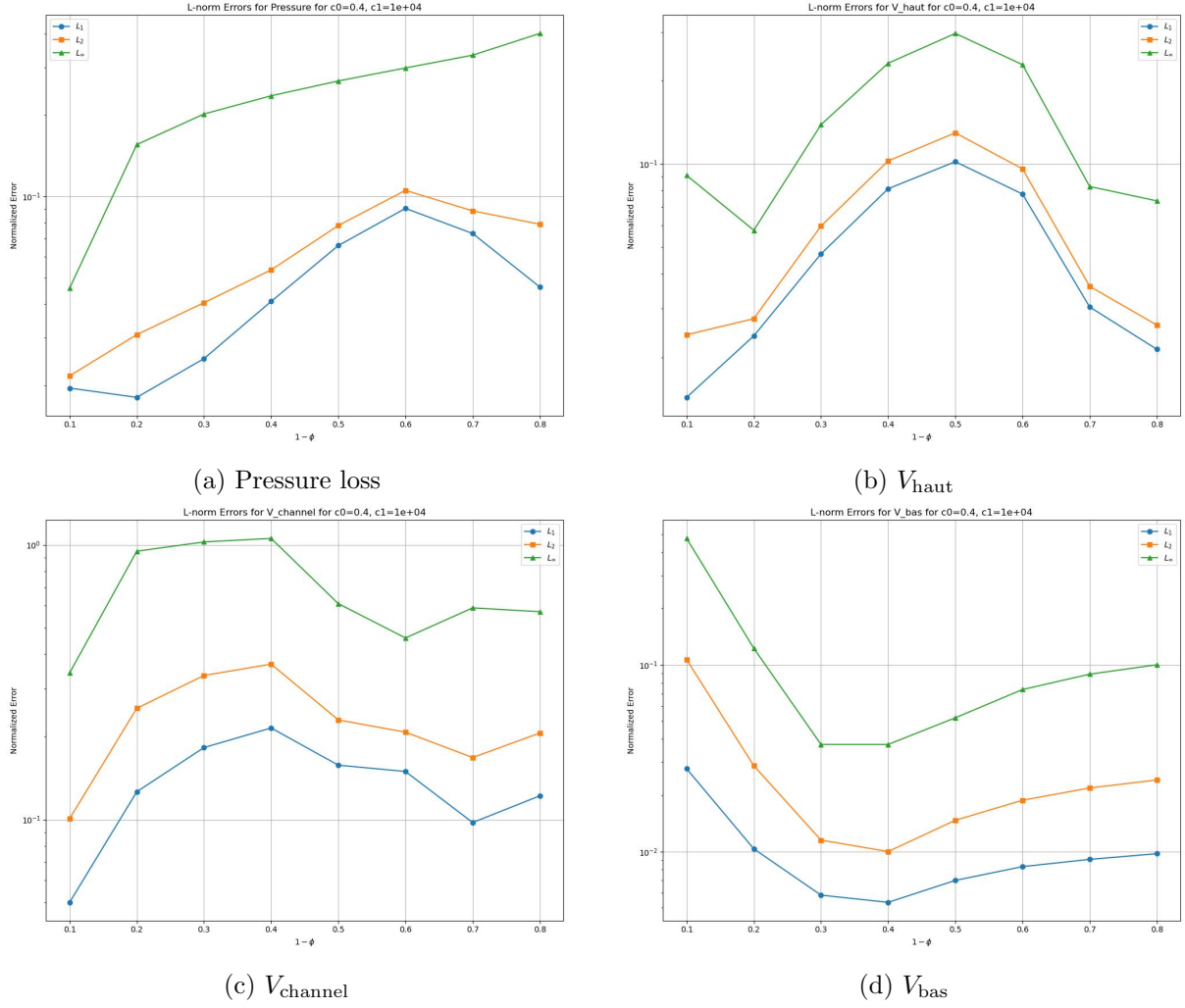


Figure 2.7: L -norm errors for the probes in Figure 2.6 for $\phi \in [0.2, 0.9]$ using $c_0 = 0.4$, $c_1 = 10^4$.

We note that the pressure loss shows a very good upper bound of about 2%, which makes this setup confidently *physical*, as minimizing the pressure loss is one of the key validation criteria we generally look for.

Overall, there is good agreement between the model and the reference cases across all probes. Looking at L_1 and L_2 , the error generally remains below 0.1 (10%), with V_{bas} showing the smallest error. This aligns with expectations, since V_{bas} lies outside the porous region and is therefore less influenced by turbulence. On the other hand, V_{channel} shows slightly higher errors in the range of 0.1–0.3, which is reasonable given that the flow inside the channel between the two porous regions is highly chaotic and therefore harder to capture accurately.

We also observe a tendency for the errors to increase around $\phi = 0.5$ on all probes. This can be explained by the randomness in the positioning of the solid portion within the porous medium: at $\phi = 0.5$ we have a roughly 50/50 fluid–solid distribution, leaving more “room” for the solid regions to be spread irregularly in the model. In contrast, for ϕ values closer to 0 or 1, the solid/fluid zones are more confined and structured, leading to more stable results.

2.3 Coefficient de frottement

Using the **équation de Darcy–Weisbach**, one can express the pressure loss as:

$$\Delta P = \Lambda \frac{L}{D_h} \frac{\rho v^2}{2}$$

where v is the vertical velocity profile, taken at the same position as the pressure loss, D_h is the hydraulic diameter of the channel, and L is the channel length.

We can compute Λ using both the reference and the model, and possibly compare it to experimental values in the future.

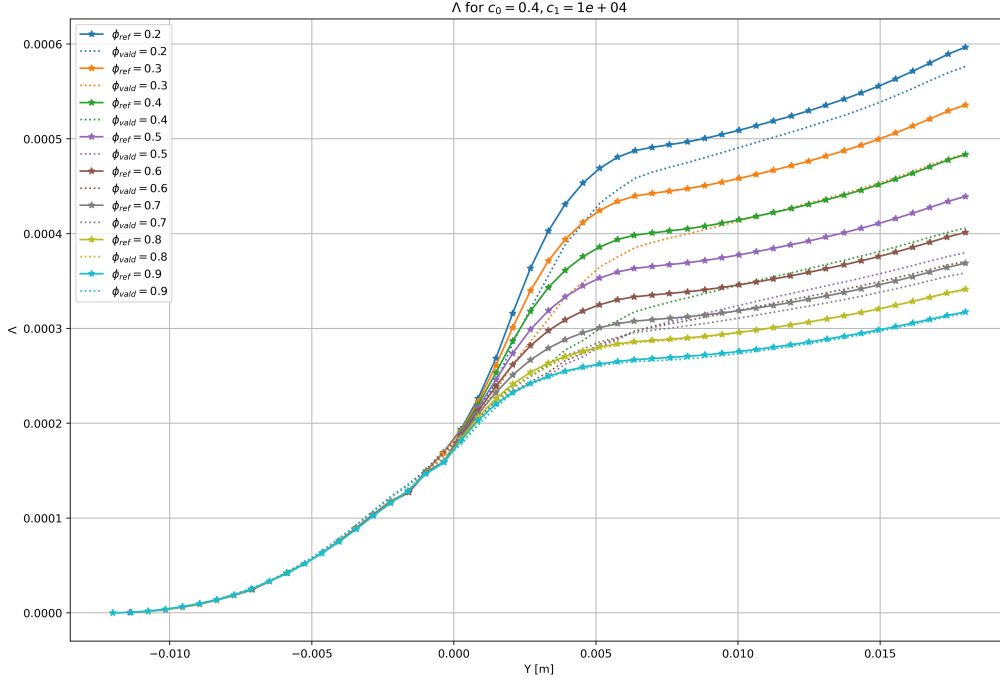


Figure 2.8: Friction coefficient for $\phi \in [0.2, 0.9]$ using $c_0 = 0.4$, $c_1 = 10^4$.

2.4 Summary

The objective for this case was to identify (c_0, c_1) values that provide the best agreement between the porous model and the reference case over a wide range of porosities. While optimizing for pressure loss, the pair:

$$c_0 = 0.4, \quad c_1 = 10^4$$

was found to offer the best compromise across all error norms and probes.

With these values, the model reproduces the reference pressure loss with an error consistently below 10%, while maintaining reasonable accuracy for velocity profiles (average deviation $\approx 11\%$ compared to probe-specific optima). The parameters are stable across $\phi \in [0.2, 0.9]$ and avoid unphysical velocity peaks at the porous interface (when examining V_{visit}).

In practise, this means we can fix (c_0, c_1) at these values for future simulations of similar geometries. However, if one is interested in absolutely reproducing the velocity probes, one should experiment with lower values for c_0 closer to possibly $\approx 10^{-6}$. The next step will be to verify the use of these parameters on additional cases.

Case 2

This case is inspired by Case 1. Here, the goal is to check whether the parameters found in Case 1 are still valid for a slightly modified geometry. The setup is again a vertical channel with two porous zones on the left and right, similar to Case 1, but with an added *transitional region* above and below the porous zone. These transitional zones are purely for mesh refinement purposes, smoothing the transition between the porous and non-porous regions.

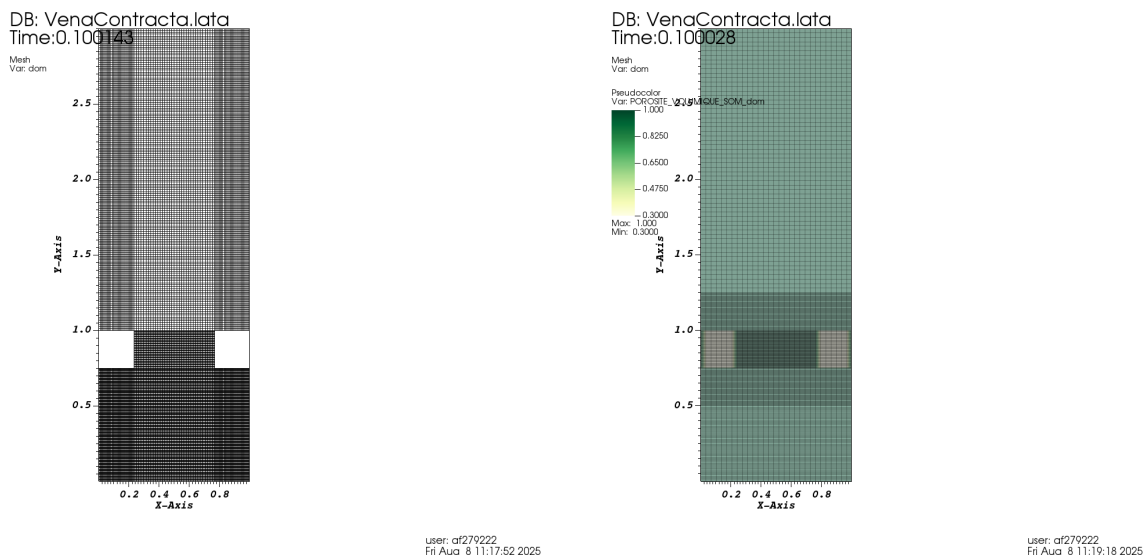


Figure 3.1: Meshes for Case 2: reference case (left) and validation case (right).

Following the same strategy as in Case 1, I first tried running the simulations with the same (c_0, c_1) values from before. However, the results were not promising:

- $\phi = 0.1$ caused the code to diverge ("Equation pbTransport_K_Eps is diverging"). $\phi = 0.2$ took a very longer to run, so I decided to avoid it by setting $\phi > 0.2$.
- Using the same $c_0 = 0.4$ and $c_1 = 10^4$, *with the same value for longueur* as in Case 1, gave large pressure losses and no flow recirculation around the porous regions in the model (unlike the reference case).
- Based on this, I decided to find new parameters for this case using the same general strategy as in Case 1 with the same initial velocity 2.7.

3.1 Finding c_1

As before, I fixed c_0 close to zero ($c_0 = 10^{-6}$) and optimized for pressure loss. I ran the model with c_1 values ranging from 10^2 up to 10^8 , refining the range until I found the “good zone,” which was roughly between 5×10^4 and 10^5 .

To save computation time, I again used only three porosities: $\phi = 0.3, 0.5, 0.9$. After some trial and error, the optimal c_1 was around 6×10^4 (Figure 3.2), giving an error of about 15% (note that this is the error between the model and the reference not the actual pressure loss which is less than 2% as we will see later).

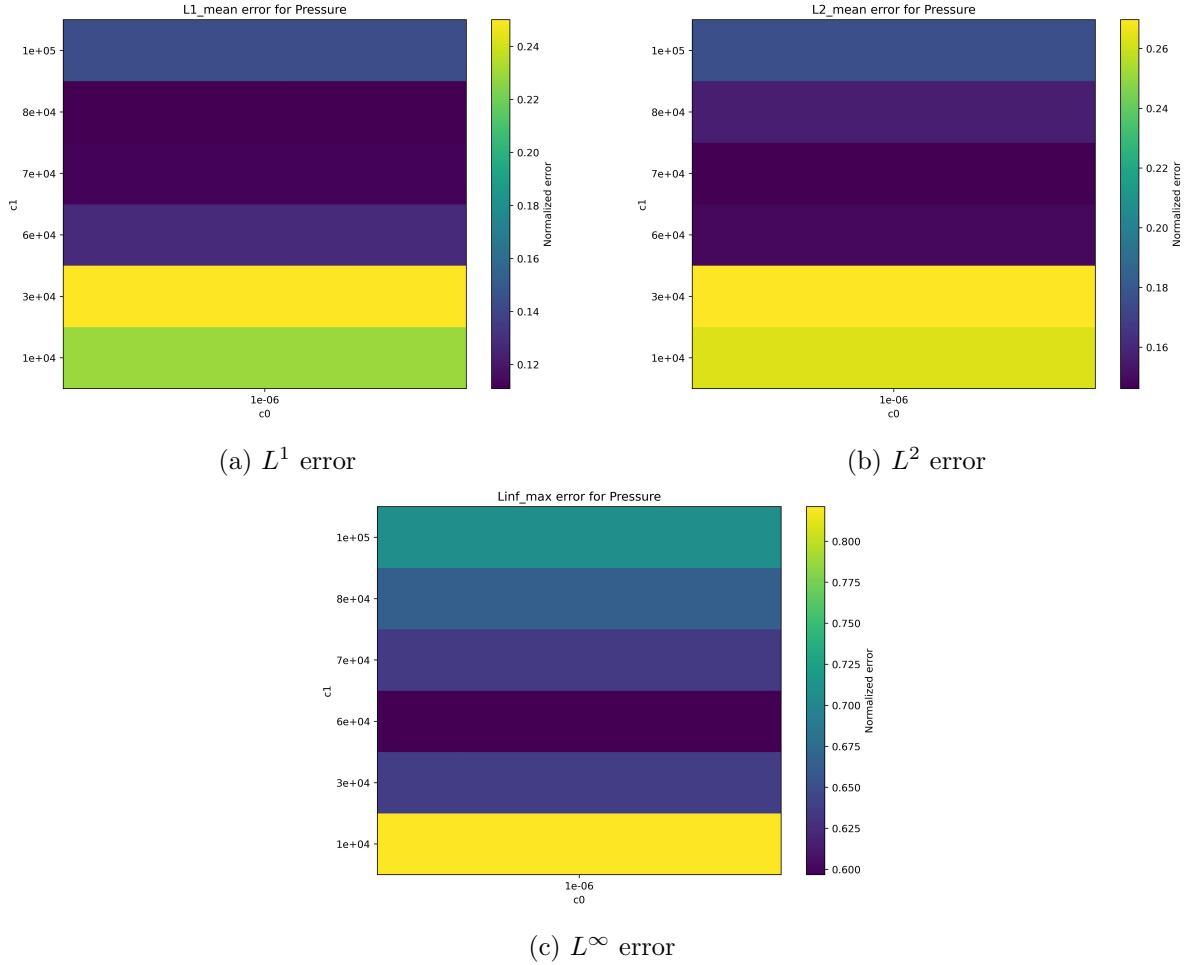


Figure 3.2: Case 2: Pressure loss error distribution heatmaps for c_1 between 10^4 and 10^5 : (a) L^1 , (b) L^2 , (c) L^∞ .

3.2 Finding c_0

Using $c_1 = 6 \times 10^4$, I searched for a good c_0 value (mostly in the $[0, 1]$ range). I tested:

$$c_0 = [10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0]$$

Interestingly, all probes (pressure and velocity) showed similar behavior: the best value was always around 10^{-6} , the smallest tested c_0 . This is unlike Case 1, where the best c_0 for pressure was 0.4 and for velocity was 10^{-6} . Here, both metrics agree on the same range.

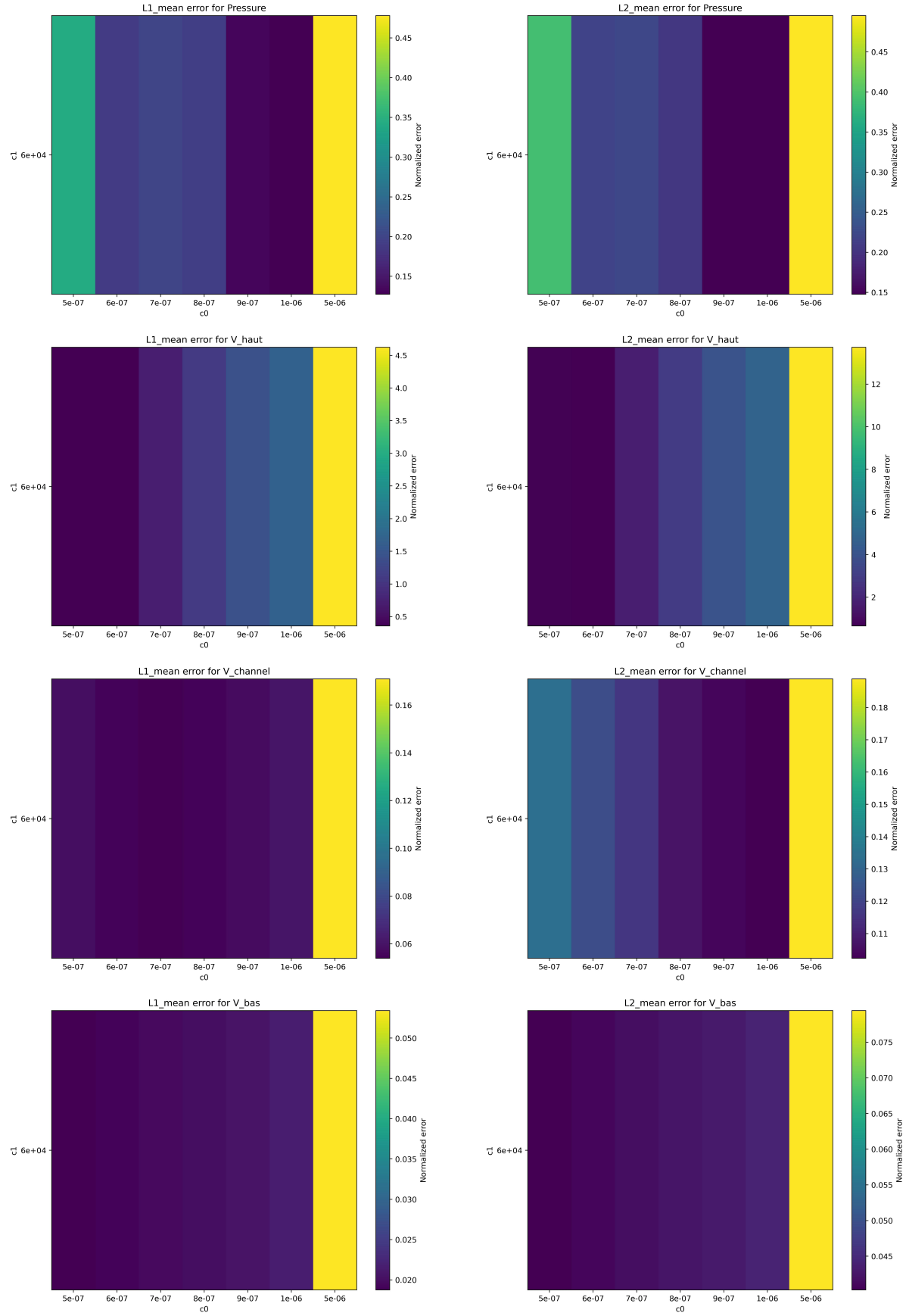


Figure 3.3: Case 2: error distribution heatmaps for various probes for c_0 between 5×10^{-7} and 5×10^{-6} .

Figure 3.3 shows these results. We note here a remarkably low error for `V_channel`, around 6% for L_1 and 11% for L_2 ; compared to Case 1 (about 15% and 30% respectively). As expected, `V_bas` shows a very small error of roughly 2%, while the pressure loss has a moderate error of about 15%. The largest error is observed for `V_haut`.

Taking these results together, and after visually inspecting the velocity and pressure profiles in VisIt, $c_0 = 10^{-6}$ appears to be a very good choice for this case. More generally, a range of 5×10^{-7} to 5×10^{-6} for c_0 and 5×10^4 to 10^5 for c_1 seems reasonable.

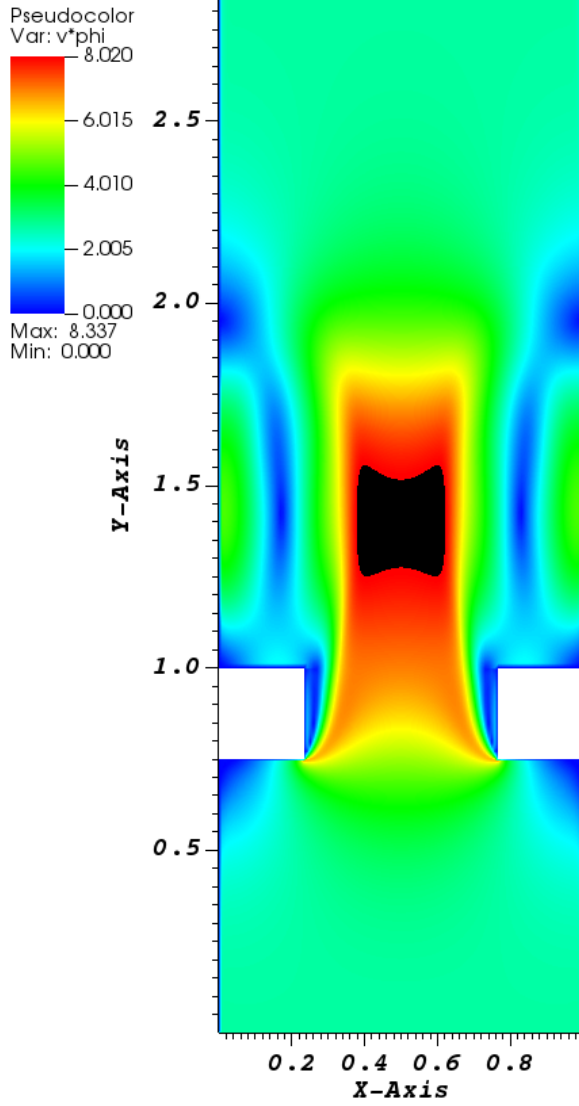
3.3 Full analysis using $c_0 = 10^{-6}$ and $c_1 = 6 \times 10^4$

Now that we have identified good parameter values, the next step is to refine the mesh, use a wider range of porosities (0.3–0.9), and test multiple inlet velocities ($v_{\text{in}} = 0.01, 2.7, 10, 100$).

For this setup I increased t_{max} to 0.5 s to allow the system more time to evolve, which is especially important for lower values of v_{in} . The mesh size was about 46,000 cells for the reference case and about 85,000 for the model. Simulation runtimes ranged from a minute to several hours, with low ϕ and high velocity cases taking the longest.

The results for $v_{\text{in}} = 2.7$ and $\phi = 0.3$ are shown in Figure 3.4 (velocity $\times\phi$) and Figure 3.5 (pressure). Both profiles are plotted for the reference (left) and the model (right), with slight adjustments to min/max scales for clearer visual comparison.

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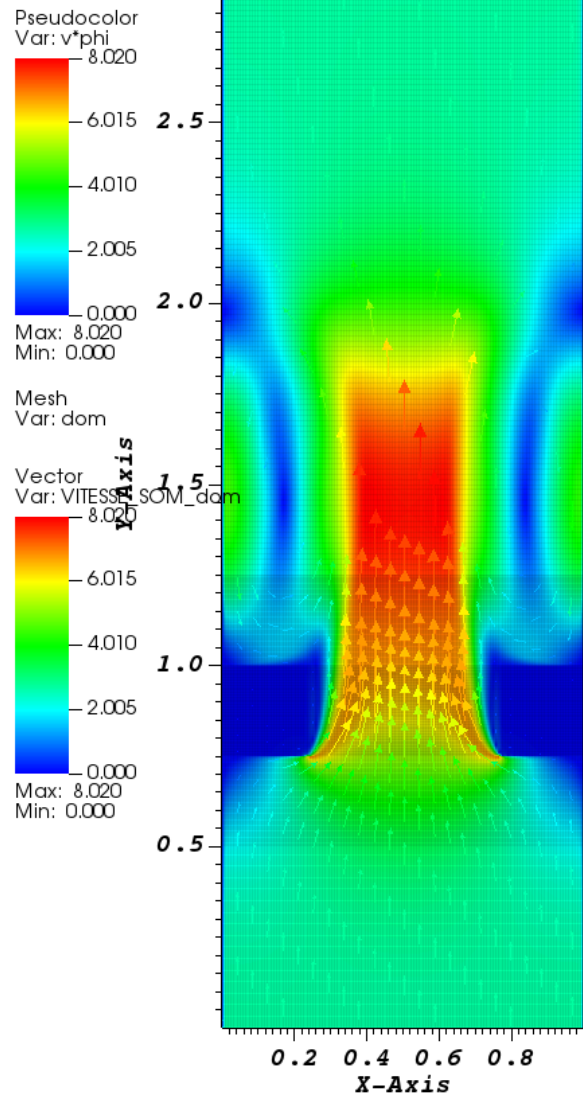


Figure 3.4: Velocity $\times \phi$ profile for $v_{in} = 2.7$, reference (left) and model (right).

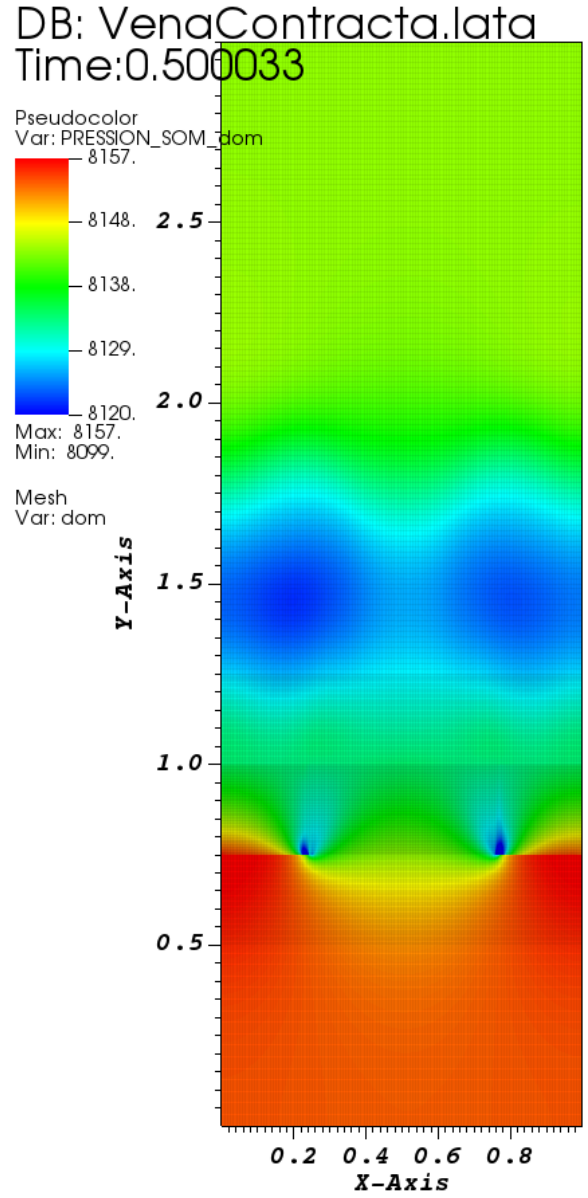
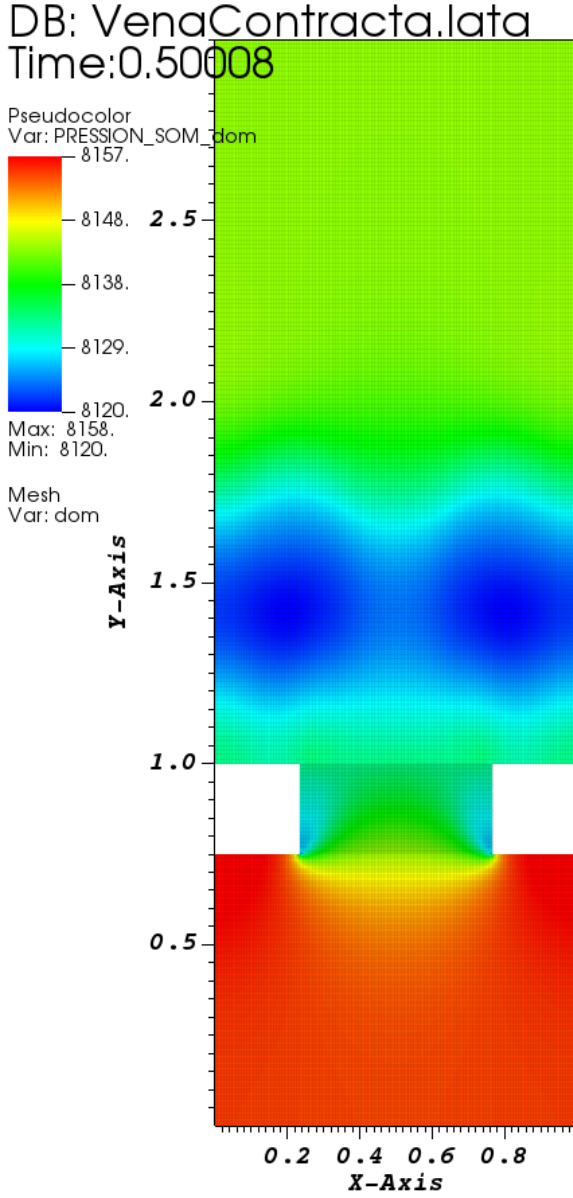


Figure 3.5: Pressure profile for $v_{in} = 2.7$, reference (left) and model (right).

We see excellent agreement between the two setups in the velocity $\times\phi$ profile, especially inside and around the channel, including the recirculation above the porous region. Similarly, the pressure profile across the domain is well reproduced, with the low-pressure region at the porous interface also captured in the model.

As in Case 1, we now examine pressure loss (vertical cut at $x = 0.5$), V_{haut} (horizontal cut just above the channel), $V_{channel}$ (inside the channel), and V_{bas} (bottom). Figures ??-?? show the L -norms and corresponding profiles for all four inlet velocities.

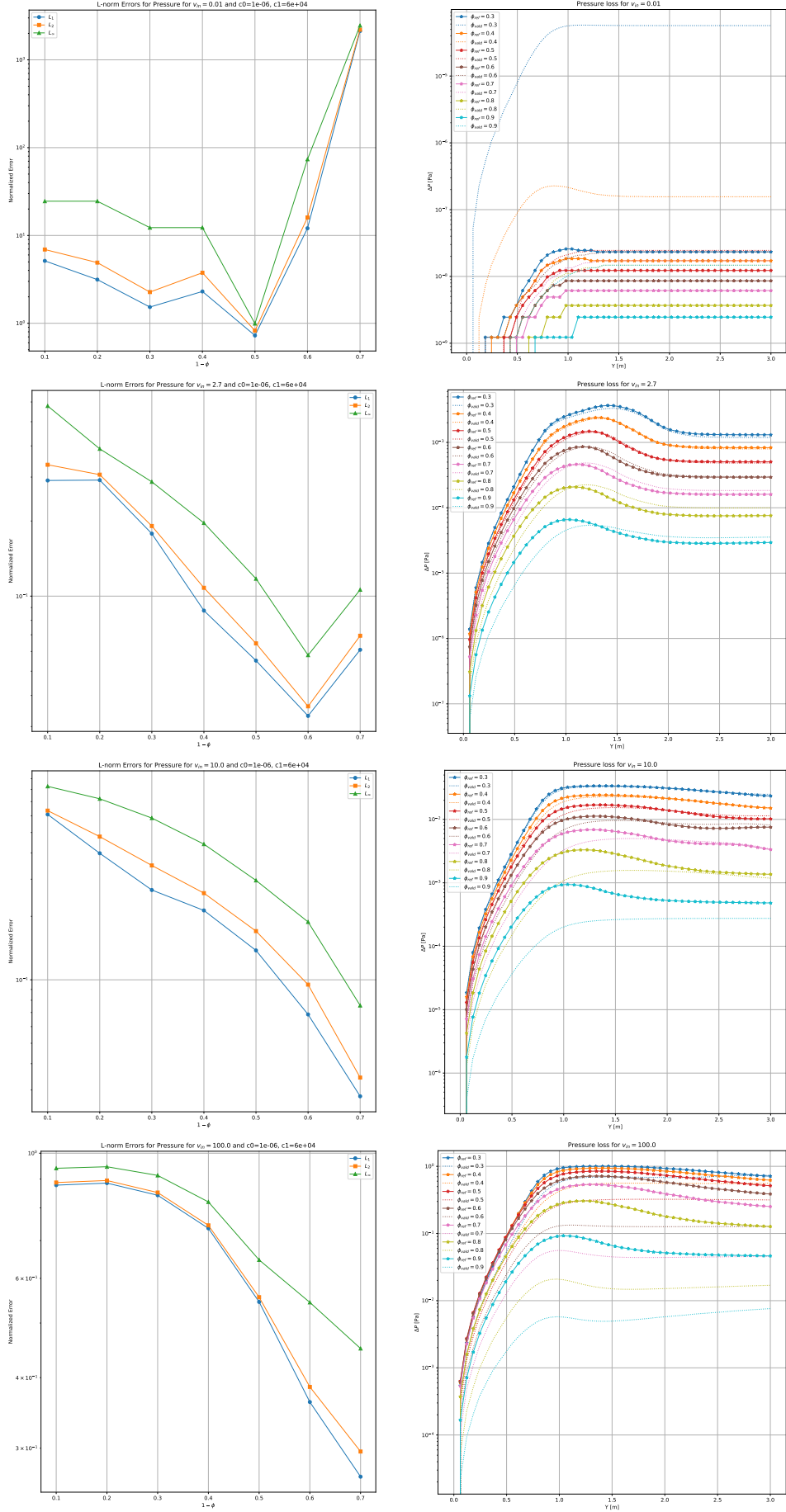


Figure 3.6: Pressure loss: L -norms (left) and profiles (right) for different v_{in} .

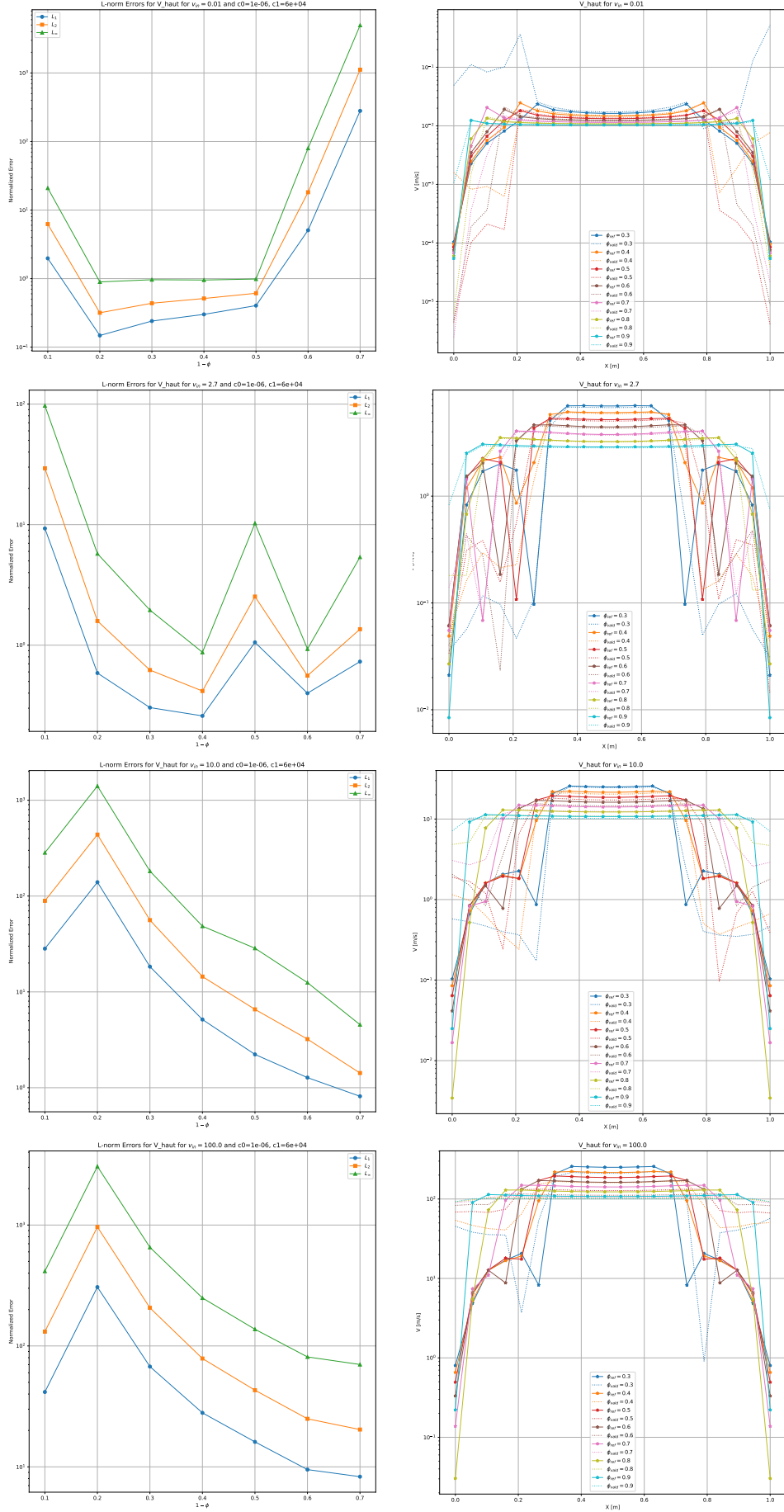


Figure 3.7: V_{haut} : L -norms (left) and profiles (right) for different v_{in} .

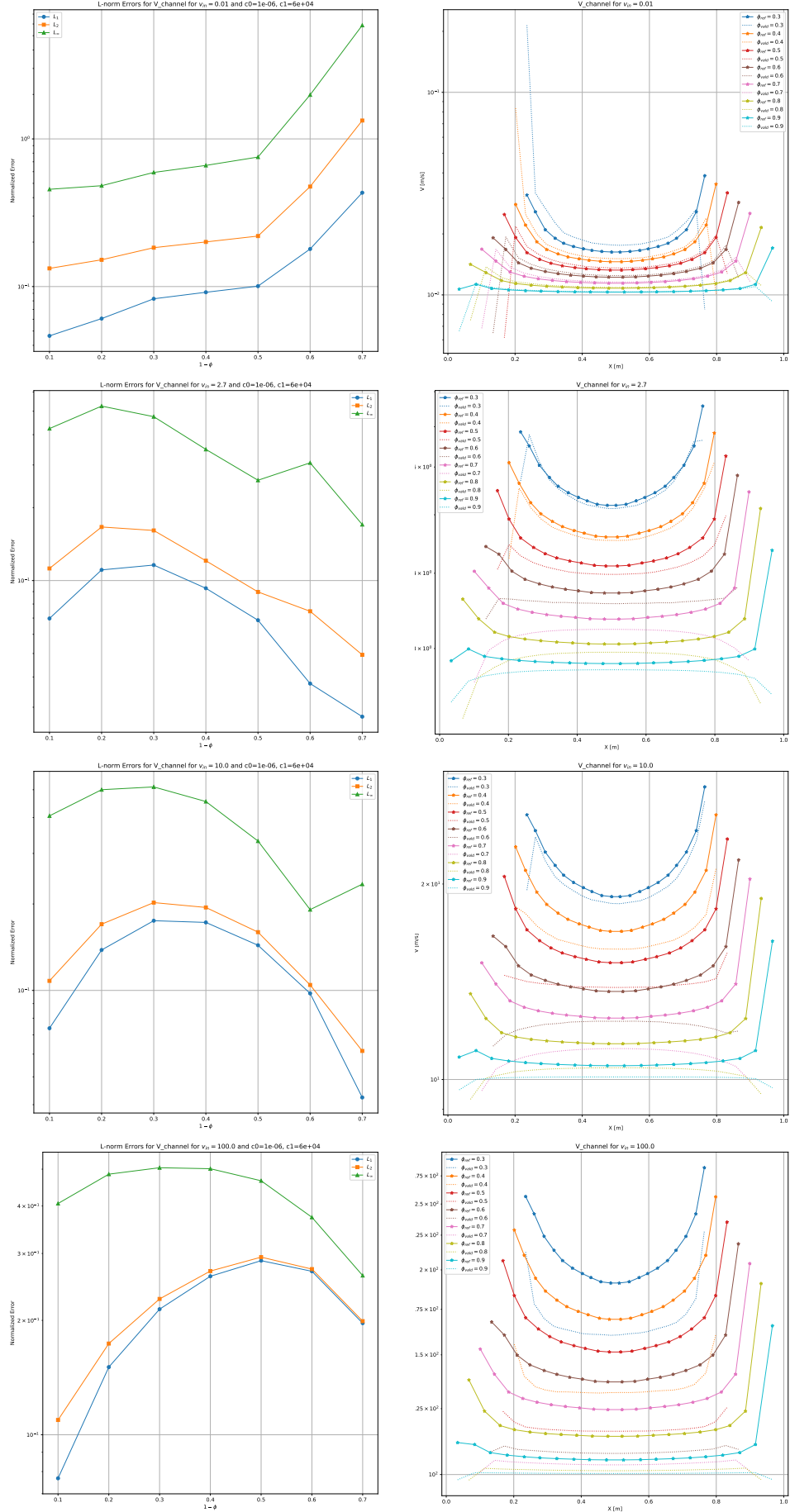


Figure 3.8: V_{channel} : L -norms (left) and profiles (right) for different v_{in} .

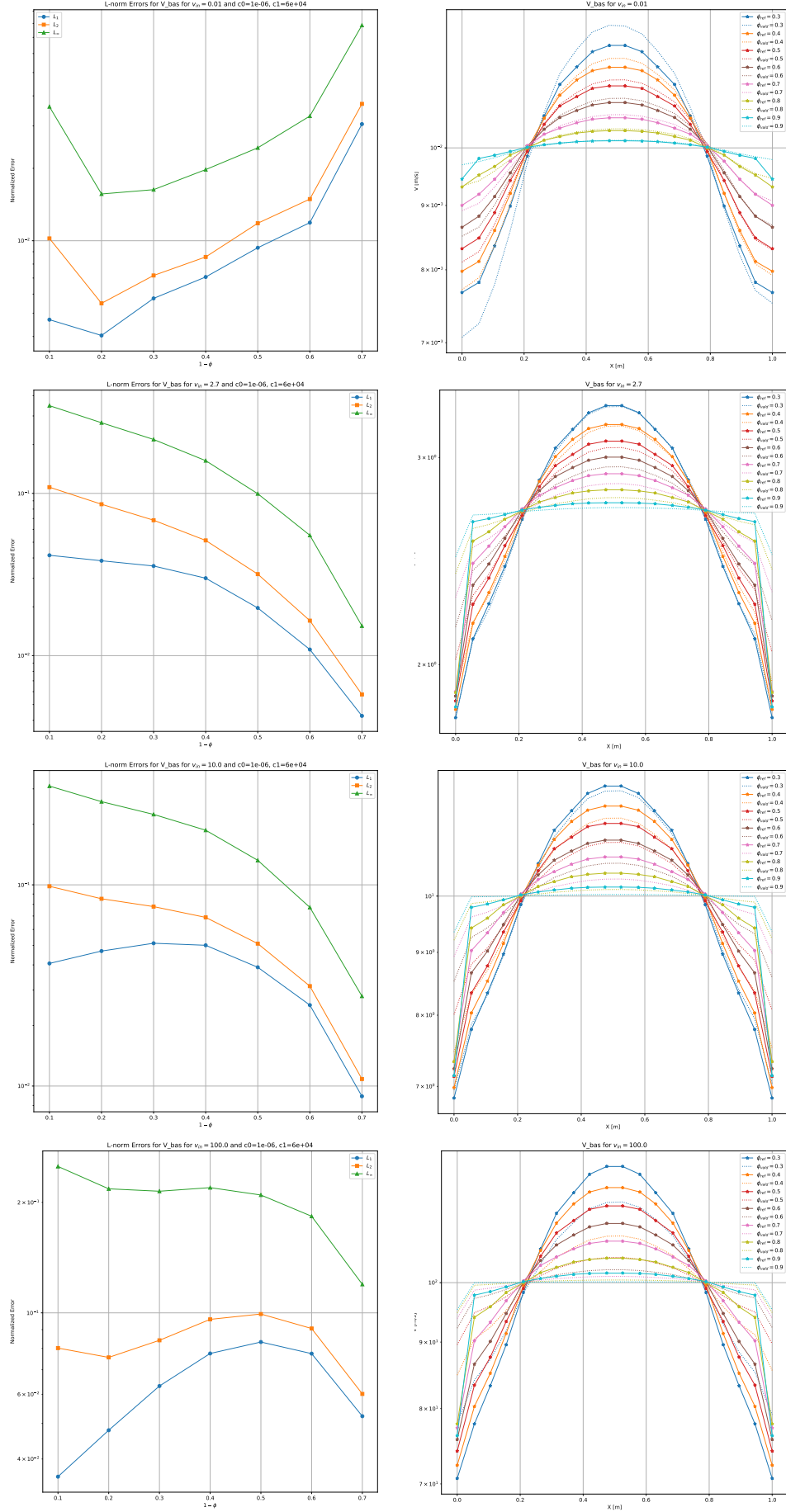


Figure 3.9: V_{bas} : L -norms (left) and profiles (right) for different v_{in} .

Interpretation. For $v_{\text{in}} = 0.01$, nothing significant happens since $t_{\text{max}} = 0.5$ s is too short; at least $t_{\text{max}} = 50$ s would be needed to see flow development. For $v_{\text{in}} = 2.7$, we observe very strong agreement in all variables with low error norms (as expected!), since (c_0, c_1) were calibrated at this velocity. For higher v_{in} , errors increase sharply, especially for V_{haut} .

Closer look of the full velocity fields in VisIt revealed an asymmetry at $v_{\text{in}} = 10$ (Figure 3.10). To test if this asymmetry is intrinsic or depended on the v_{in} , I reran the $v_{\text{in}} = 2.7$ case with $t_{\text{max}} = 2$ s to allow the system to evolve and reach a similar state to that of $v_{\text{in}} = 10$. As shown in Figure 3.11, the asymmetry disappeared, confirming that the effect is linked to high v_{in} rather than the (c_0, c_1) values, and may relate to how velocity direction is handled in the C++ code.

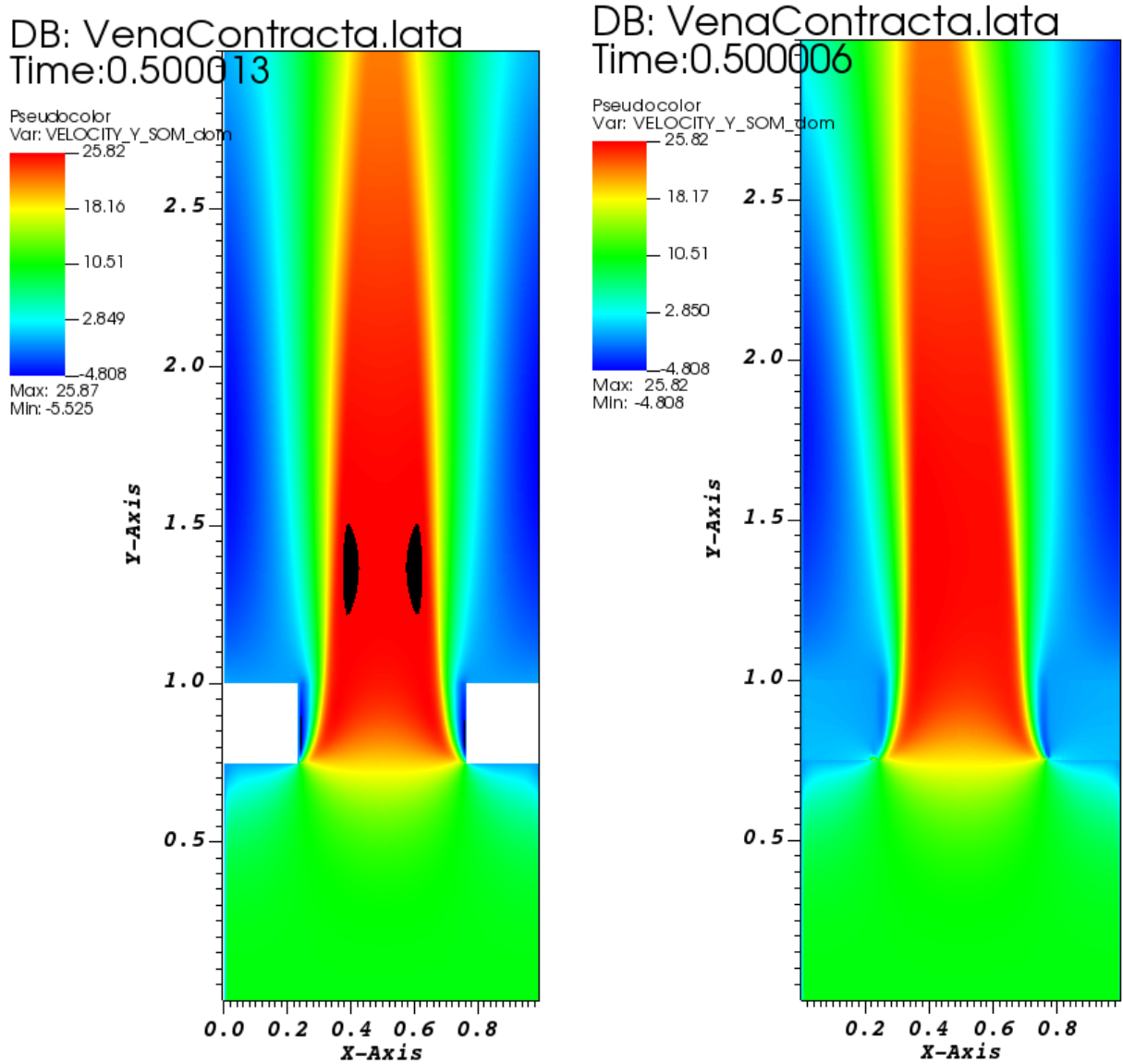


Figure 3.10: Velocity profile for $v_{\text{in}} = 10$: reference (left) and model (right). Asymmetry visible in the model.

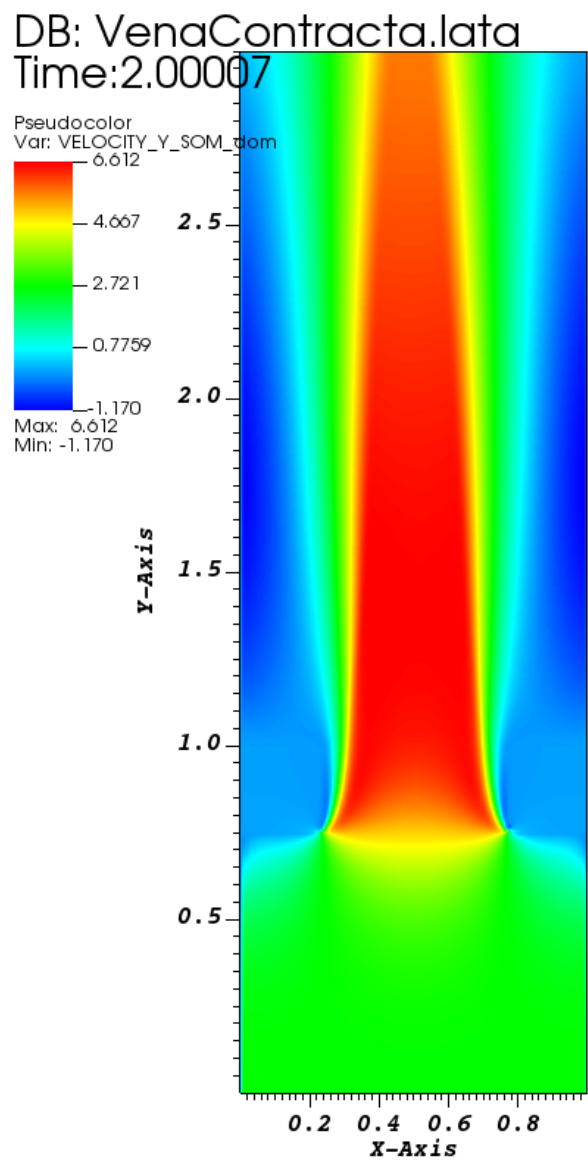
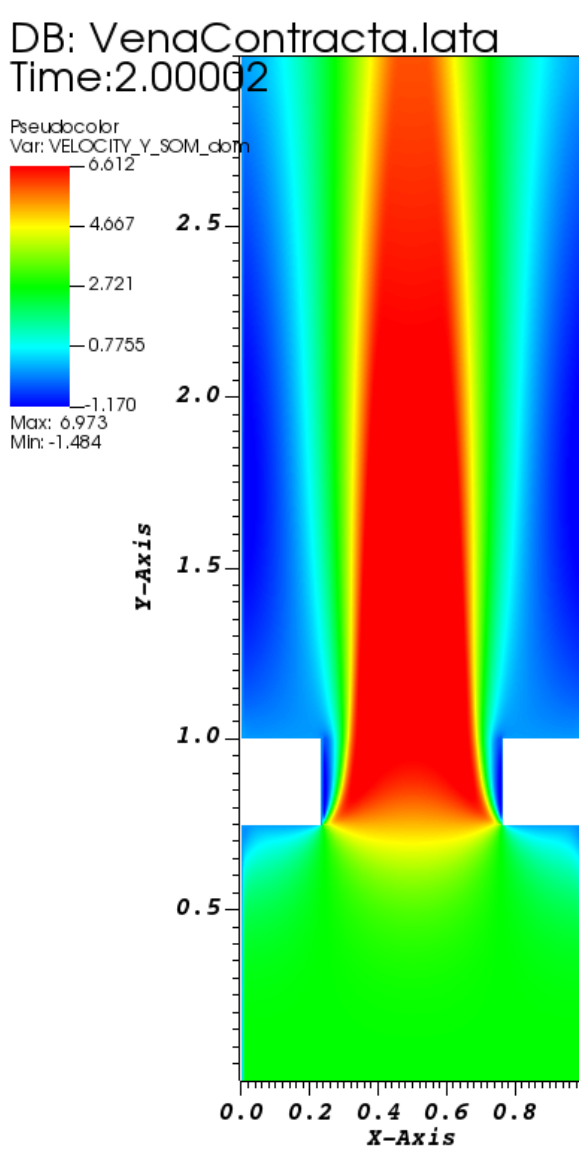


Figure 3.11: Velocity profile for $v_{in} = 2.7$, $t_{max} = 2$ s: reference (left) and model (right). No asymmetry.

3.4 Coefficient de frottement

Finally, Figure 3.12 shows the friction coefficient for $v_{in} = 2.7$. This is included mainly for documentation and may be compared against experimental data in future work.

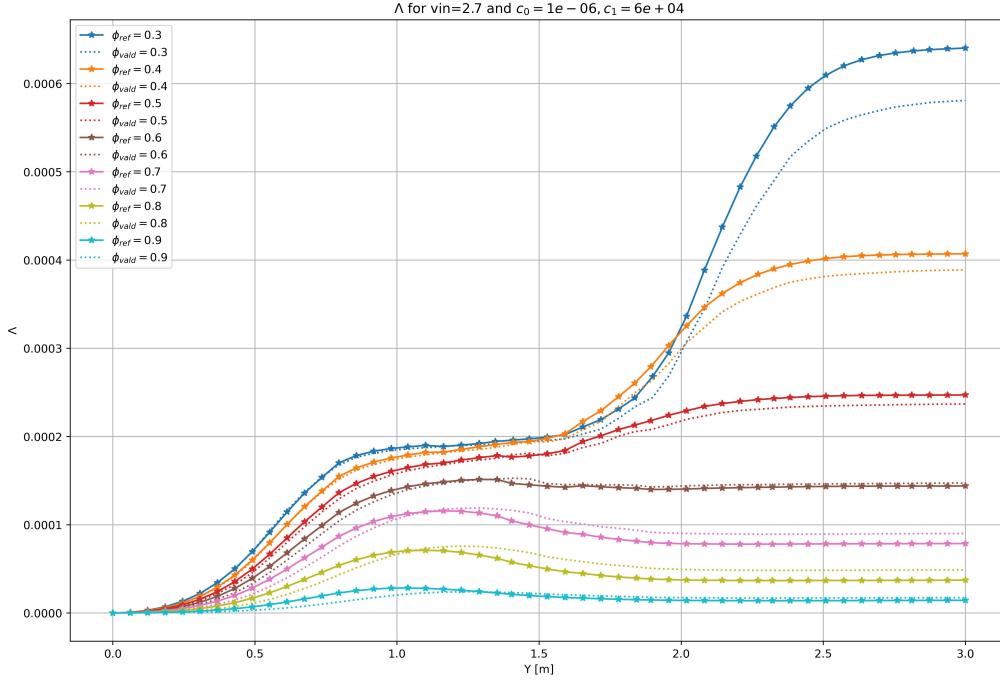


Figure 3.12: Case 2: Friction coefficient for $\phi \in [0.3, 0.9]$ using $c_0 = 10^{-6}$, $c_1 = 6 \times 10^4$.

3.5 Summary

- $(c_0, c_1) = (10^{-6}, 6 \times 10^4)$, with ranges $c_0 \in [5 \times 10^{-7}, 5 \times 10^{-6}]$ and $c_1 \in [5 \times 10^4, 10^5]$, gives promising results, especially at $v_{in} = 2.7$.
- Asymmetry appears at higher $v_{in} > 10$; likely linked to velocity direction handling in the C++ code.

Chapter 4

Conclusion

If I were to recommend a pair of values for (c_0, c_1) to use in general with this model, based on both Case 1 and Case 2, then c_1 is quite clear: in both cases we ended up in the same ballpark, around 10^4 – 10^5 .

For c_0 it's a bit trickier: in Case 1 we picked 0.4, while in Case 2 the best choice was 10^{-6} . But actually, even in Case 1 using 10^{-6} gave better velocity results, and the pressure loss error was only 10–20% worse than with $c_0 = 0.4$ (see L^1 mean error in Figure 2.2). So all in all, it's more practical to stick with the smaller value.

One limitation we observed is that at higher inlet velocities ($v_{in} > 10$), the model starts showing asymmetry in the velocity field compared to the reference. This does not appear at moderate velocities ($v_{in} = 2.7$), and seems linked more to the treatment of velocity direction in the code than to the choice of (c_0, c_1) . we need modify the code, rerun `assymetry.ipynb`, and verify in VisIt.

My recommendation is therefore:

$$c_0 = 10^{-6}, \quad c_1 = 6 \times 10^4$$

with a tolerance of about one order of magnitude around each. That should be a good and practical starting point for any future case with this type of geometry.